Andrea Caputo

Theoretical particle physics, astrophysics, and cosmology
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Publications (INSPIRE) | Research software (GitHub) | ORCID

Academic appointments

Staff Member, Theoretical Physics Department — Dec 2025–present
CERN, Geneva, Switzerland

Associate Professor of Theoretical Physics — Nov 2024–present
Sapienza University of Rome, Italy

Senior Research Fellow — Nov 2022–Nov 2025
CERN, Geneva, Switzerland

Postdoctoral Researcher — Oct 2020–Oct 2022
Tel Aviv University and Weizmann Institute of Science, Israel

Education

Ph.D. in Theoretical Physics — Sep 2016–Sep 2020
University of Valencia, Spain; *cum laude*.
Advisor: Prof. Pilar Hernández.

M.Sc. in Theoretical Physics — Sep 2014–Jul 2016
Sapienza University of Rome, Italy.

B.Sc. in Physics — 2011–2014
University of Calabria, Italy.

Research funding and leadership

ERC Starting Grant, AstroDarkLS — Awarded 2023
Principal Investigator; EUR 1.5 million research program.

- Investigate dark matter and new particles through astrophysical and cosmological probes, including supernovae and the Lyman-α forest.
- Direct research strategy, grant execution, hiring, mentoring, and collaboration across institutions.

Honors and fellowships

2025	The Benoziyo Endowment Visiting Professor, Weizmann Institute of Science
2023	IUPAP Early Career Scientist Prize in Astroparticle Physics.
2023	First place, CNRS researchers competition, section 02 CRCN (declined).
2022	Marie Skłodowska-Curie Individual Fellowship (awarded; declined).
2022	Azrieli International Fellowship.
2020	Israel Academy of Science Fellowship.
2018	Giuliano Preparata Prize, Italian Physical Society.

Selected publications

The full publication list is supplied as a separate document and is available on INSPIRE, here I just list a few highlights.

1. **Superradiance constraints from GW231123.**
   A. Caputo, G. Franciolini, and S. J. Witte.
   Physical Review D **113**, 083031 (2026).

2. **Evolution of Binaries Under Stochastic Perturbations.**
   A. Caputo, G. M. Tomaselly and Chris Hamilton.
   Accepted in Physical Review D

3. **Influence of tides and self-gravity on ultralight dark matter bounds from dwarf galaxies.**
   A. Caputo and L. Teodori.
   Physical Review D **114**, 063013 (2026).

4. **Introducing the DREAMS Project: DaRk mattEr and Astrophysics with Machine Learning and Simulations.**
   J. C. Rose et al. (including A. Caputo).
   Astrophysical Journal **982**, 68 (2025).

5. **Pulsar Nulling and Vacuum Radio Emission from Axion Clouds.**
   A. Caputo, S. J. Witte, A. A. Philippov, and T. Jacobson.
   Physical Review Letters **133**, 161001 (2024).

6. **Low-Energy Supernovae Severely Constrain Radiative Particle Decays.**
   A. Caputo, H.-T. Janka, G. Raffelt, and E. Vitagliano.
   Physical Review Letters **128**, 221103 (2022).

Teaching and research supervision

Sapienza University of Rome Nov 2024–present

- Teaching in Mathematical Methods for Physicists.
- Teaching in Classical Mechanics.
- Supervising at the moment 2 PhD students, 2 postdocs and 3 undergraduate students. I have co-supervised in the past other three PhD students (between Tel Aviv and Sapienza University), and 5 undergraduate students (Sapienza University and CERN internships).

Languages

Italian (native); English and Spanish (fluent); French and Hebrew (basic).

References

Pilar Hernández (Ph.D. supervisor)
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